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Dental Approach to Barodontalgia in Scuba Divers: A Narrative Review

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Abstract

Background. Barodontalgia is dental or oral pain triggered by changes in ambient pressure. In scuba divers, this condition is clinically relevant because it may indicate previously undetected dental pathology and may affect comfort, concentration and underwater safety.

Aim. The aim of this narrative review is to summarize current knowledge on barodontalgia in scuba divers from a dental perspective, with emphasis on risk factors, clinical presentation, diagnosis, management and prevention.

Material and methods. A narrative literature review was conducted using PubMed/MEDLINE, Scopus, Web of Science and Google Scholar. The search strategy included combinations of the terms “barodontalgia”, “tooth squeeze”, “dental barotrauma”, “odontocrexia”, “scuba diving”, “divers”, “diving dentistry”, “hyperbaric dentistry”, “dental pain”, “dental restorations” and “endodontic treatment”. Review articles, observational studies, case reports and in vitro studies related to pressure-induced dental and orofacial complications were included.

Results. The literature indicates that barodontalgia is usually associated with pre-existing oral pathology, including dental caries, defective restorations, pulpitis, pulp necrosis, apical periodontitis, periodontal disease, residual air spaces, impacted teeth and maxillary sinus disorders. Dental barotrauma may additionally involve fracture or dislodgement of restorations, crowns or prosthetic components. Diagnosis can be difficult because symptoms may occur only during descent or ascent and may not be reproducible during routine dental examination.

Conclusions. Dentists should consider barodontalgia in divers who report pressure-related tooth pain. A practical dental approach should include elimination of active odontogenic disease, assessment of restorations and endodontically treated teeth, differential diagnosis of sinus- and temporomandibular-related pain, and preventive counselling before diving activity.

Keywords: barodontalgia, scuba diving, dental barotrauma, diving dentistry, tooth squeeze, odontocrexia, dental prevention

1. Introduction

Scuba diving exposes the human body to repeated changes in ambient pressure. These changes are usually well tolerated by trained divers; however, air-containing spaces and tissues that cannot equalize pressure may become symptomatic or injured. In the head and neck region, pressure-related disorders are commonly discussed in relation to the middle ear and paranasal sinuses, but the oral cavity and dental structures may also be involved. From a dental perspective, one of the most important pressure-related manifestations is barodontalgia, commonly described as tooth pain induced by barometric pressure changes (Rauch, 1985; Robichaud & McNally, 2005; Zadik, 2009).

Barodontalgia is not usually regarded as an independent disease entity. Rather, it is often interpreted as a clinical symptom revealing pre-existing pathology that may be asymptomatic under normal atmospheric conditions. Potential sources include caries, defective restorations, pulpitis, pulp necrosis, periapical inflammation, periodontal disease, impacted teeth and other odontogenic or non-odontogenic conditions (Robichaud & McNally, 2005; Zadik, 2009; Zadik & Drucker, 2011). This makes the condition particularly relevant for general dental practitioners, because a diver who reports pain only during diving may not present with obvious symptoms during a routine office visit.

The dental implications of diving have been reviewed previously, including broader discussions of barotrauma, barosinusitis, temporomandibular disorders and mouthpiece-related complaints (Zadik & Drucker, 2011; Livingstone & Lange, 2018; Sionek-Wręga & Wręga, 2024). More recent literature has also emphasized barodontalgia in military and recreational divers, including the need for preventive dental protocols and interdisciplinary cooperation between dentists, diving physicians and diving instructors (García-Torres & Segura-Egea, 2024; Alolaiwi et al., 2025). Nevertheless, many available papers either focus on aviation dentistry, general diving medicine or broad orofacial complaints. A concise dental approach directed specifically at barodontalgia in scuba divers remains useful for clinicians who may encounter divers in everyday practice.

The practical importance of the topic is supported by survey studies among recreational, civilian, occupational and military divers. Reported rates vary substantially due to differences in population, methodology and definitions, but studies consistently show that dental pain, restoration damage and orofacial discomfort occur in a clinically relevant subgroup of divers (Jagger et al., 2009; Gunepin et al., 2015; Gunepin et al., 2016; Ranna et al., 2016; Alwohaibi et al., 2020; Onose et al., 2020; Moyaux et al., 2022; Gougeon et al., 2022; Sumen et al., 2023). This variation also indicates that the evidence base is heterogeneous and that recommendations

are often based on observational data, case reports, laboratory studies and expert opinion rather than high-level controlled trials.

For dentists, the central clinical question is not only how barodontalgia develops, but how it should be anticipated, diagnosed and managed. Diving candidates may ask whether they can dive after restorative treatment, root canal treatment, extraction, orthodontic procedures or prosthetic rehabilitation. Conversely, experienced divers may present with pain that appears only during descent or ascent. In such situations, the dentist must consider odontogenic pathology, sinus-related referred pain, temporomandibular joint disorders and pressure-related failure of dental materials.

Research Objective. The objective of this narrative review is to summarize current knowledge on barodontalgia in scuba divers from a dental perspective, with particular emphasis on dental risk factors, diagnosis, management and prevention.

Research Problems. The review addresses three practical questions: (1) What dental conditions and restorative factors are associated with barodontalgia and dental barotrauma in scuba divers? (2) How can dentists approach diagnosis and differential diagnosis when symptoms occur mainly during diving? (3) What preventive recommendations can be proposed for recreational, occupational and military divers?

Research Hypotheses. Due to the narrative review design, no statistical hypotheses were tested. The working assumptions were that barodontalgia in divers is most often associated with pre-existing dental or orofacial pathology, that prevention depends primarily on stabilization of oral health before diving activity, and that dentists have an important role in identifying pressure-related risk factors before and after diving exposure.

2. Research materials and methods

2.1. Study design and literature search strategy

This study was conducted as a narrative review. The review was designed to provide a clinically oriented synthesis for dental practitioners rather than a formal systematic review or meta-analysis. A structured literature search was performed using PubMed/MEDLINE, Scopus, Web of Science and Google Scholar. The search was updated in July 2026.

The search strategy included the following keywords and their combinations: “barodontalgia”, “tooth squeeze”, “dental barotrauma”, “odontocrexia”, “scuba diving”, “divers”, “diving dentistry”, “hyperbaric dentistry”, “dental pain”, “orofacial pain”, “dental restorations”, “endodontic treatment”, “pressure changes” and “barotrauma”. Reference lists of relevant review articles and original studies were screened manually to identify additional publications.

Particular attention was paid to papers with practical implications for dental examination, restorative dentistry, endodontics, prosthodontics and preventive care in divers.

2.2. Inclusion criteria and exclusion criteria

Eligible publications included narrative reviews, systematic reviews, observational studies, cross-sectional surveys, case reports and laboratory studies discussing barodontalgia, dental barotrauma or dental complications associated with pressure changes during scuba diving, hyperbaric exposure or comparable barometric conditions. Articles were considered particularly relevant if they addressed dental causes, clinical presentation, diagnosis, treatment considerations, restorative materials, endodontic factors or prevention.

Publications were excluded if they were not related to dental or orofacial manifestations, if they discussed diving injuries without relevance to oral health, if only non-medical commentary was available, or if bibliographic data were insufficient for citation. Because the aim was practical and clinically oriented, both older foundational publications and recent evidence were considered. No language restriction was planned at the search stage, but English-language sources and sources with accessible English bibliographic data were prioritized for manuscript preparation.

2.3. Data collection and qualitative synthesis

The selected literature was grouped thematically into the following categories: general concepts and definitions of barodontalgia; prevalence and risk factors among divers; restorative and prosthetic complications; endodontic considerations; temporomandibular and sinus-related differential diagnosis; and prevention. Due to substantial heterogeneity in study designs, populations, pressure-exposure protocols and outcome definitions, a qualitative synthesis was performed without meta-analysis. This approach is consistent with the practical purpose of the article and with the limited comparability of available studies.

The final narrative synthesis focused on publications that provided clinically useful information for dentists. Observational studies and surveys were used primarily to summarize prevalence patterns and reported risk factors, while laboratory studies were used to discuss possible mechanisms involving restorative materials, crowns, posts, root fillings and microleakage. Case reports were used only to illustrate clinically plausible complications and were not treated as evidence of frequency.

2.4. AI assistance

Artificial intelligence tools were used as assistive instruments for organizing the manuscript structure, improving academic English and supporting editorial consistency. The final selection of literature, interpretation of clinical content, verification of references and conclusions require human author review. AI tools were not used as a source of clinical evidence, did not replace professional judgment and did not generate original research data.

3. Research results

3.1. Definition and mechanisms of barodontalgia

Barodontalgia is usually defined as dental or oral pain caused by changes in ambient pressure. It has also been described as “tooth squeeze” and has historically been discussed in aviation, diving and hyperbaric medicine (Rauch, 1985; Zadik, 2009; Zadik, 2010). The mechanism is commonly related to pressure gradients acting on gas-filled spaces, fluid shifts or inflammatory tissues within dental or adjacent orofacial structures. Boyle's law provides the basic physical framework: when ambient pressure increases, gas volume decreases, and when pressure decreases, gas volume expands. In the presence of trapped gas, defective restorations, dentinal defects or poorly sealed spaces, such changes may produce pain or mechanical stress (Hamilton-Farrell & Bhattacharyya, 2004; Stoetzer et al., 2012).

In scuba diving, barometric changes occur rapidly during descent and ascent. A diver descending to depth is exposed to increasing ambient pressure, while ascent is associated with decreasing pressure and expansion of gas spaces. If pressure cannot be equalized, tissue damage or pain may develop. In dentistry, the relevant spaces are not always visible. They may include microscopic gaps under restorations, voids in luting cement, residual spaces in endodontic systems, carious cavities, periodontal pockets or maxillary sinus spaces adjacent to posterior teeth (Zadik & Drucker, 2011; Livingstone & Lange, 2018).

A clinically important feature is that barodontalgia is often a symptom of underlying disease rather than a disease itself. This distinction should shape dental management. The goal is not only to relieve pain, but to identify the dental or non-dental condition that becomes symptomatic under pressure stress. For this reason, barodontalgia should prompt a careful diagnostic work-up even if the patient is asymptomatic at rest (Robichaud & McNally, 2005; Zadik, 2009; García-Torres & Segura-Egea, 2024).

Dental barotrauma is related but not identical. It refers to physical damage to dental tissues, restorations or prosthetic elements caused by pressure change, with or without pain. Examples

include fracture of restored teeth, dislodgement of crowns or provisional restorations, loss of fillings and disruption of prosthetic or orthodontic components (Calder & Ramsey, 1983; Lyons et al., 1999; Gulve & Gulve, 2013a; Gougeon et al., 2022). A diver may therefore experience barodontalgia without visible dental damage, dental barotrauma without pain, or both simultaneously.

3.2. Dental causes and risk factors

The literature consistently links barodontalgia with pre-existing oral pathology. Caries, deep restorations, reversible or irreversible pulpitis, pulp necrosis, apical periodontitis, periodontal disease and pericoronitis may become symptomatic under barometric stress (Robichaud & McNally, 2005; Zadik, 2009; Zadik & Drucker, 2011). In diving, posterior and maxillary teeth are frequently discussed because of their proximity to the maxillary sinuses and the prevalence of posterior restorative treatment. However, any tooth or orofacial structure can be involved depending on the underlying condition.

Defective restorations are of particular dental relevance. Microleakage, open margins, recurrent caries, fractured restoration margins and trapped air spaces may create a situation in which pressure changes provoke pain or restoration failure. Lyons et al. (1999) reported that simulated diving pressure cycles may affect microleakage and retention of full cast crowns. Later reports and laboratory studies continued to support the idea that restorative interfaces can be vulnerable to pressure cycling, although results differ according to material, restoration design, cement and experimental protocol (Khanna, 2010; Shafigh et al., 2018; Shahnavazi et al., 2021; Ata et al., 2022).

Endodontic status is another important factor. A tooth with untreated pulp disease may become painful during pressure change, but endodontically treated teeth also require attention. Laboratory evidence suggests that the sealing ability of root canal filling techniques can differ under simulated diving conditions. Stoetzer et al. (2012) reported that warm gutta-percha obturation techniques provided a more pressure-tight seal than cold lateral condensation or Resilon-based techniques in their experimental model. Safai et al. (2019) and von See et al. (2012) further support the clinical relevance of endodontic sealing and post-retention questions under pressure exposure.

Prosthetic and orthodontic components may also be affected. Case reports describe dislodgement of provisional crowns during scuba diving, and in vitro studies have examined glass fiber posts, orthodontic brackets and luting cements under simulated pressure changes (Gulve & Gulve, 2013a; Gulve & Gulve, 2013b; Gulve & Gulve, 2022). These studies cannot

be directly translated into universal clinical rules, but they suggest that recent restorations, provisional cementation, inadequate retention and marginal defects deserve special attention before diving.

Observational studies show that reported risk is influenced not only by dental status but also by diving exposure. Variables such as diving depth, frequency of dives, occupational duration, military status and diving qualification may influence the probability of symptoms or reported dental barotrauma (Gunepin et al., 2015; Gunepin et al., 2016; Alwohaibi et al., 2020; Moyaux et al., 2022; Gougeon et al., 2022; Alolaiwi et al., 2025). This is clinically logical: repeated and deeper pressure cycling increases the opportunity for vulnerable dental structures to become symptomatic.

Table 1. Key literature relevant to the dental approach to barodontalgia in scuba divers.

Study	Design / population	Main relevance for this review
Zadik (2009)	Review	Definition, clinical features, pathogenesis and preventive concepts of barodontalgia.
Zadik & Drucker (2011)	Review	Dental implications of scuba diving, including oral barotrauma and mouthpiece-related issues.
Robichaud & McNally (2005)	Review	Barodontalgia as a differential diagnosis and clinical management issue.
Jagger et al. (2009)	Survey	Orofacial pain and odontocrexia associated with scuba diving.
Gunepin et al. (2015)	Military diver survey	Dental barotrauma in French military divers.
Gunepin et al. (2016)	Military diver study	Incidence and features of barodontalgia among military divers.
Ranna et al. (2016)	Pilot survey	Dental problems reported by recreational scuba divers.
Alwohaibi et al. (2020)	Cross-sectional study	Dental and orofacial barotrauma among Saudi military naval divers.
Onose et al. (2020)	Occupational diver study	Oral symptoms and preventive dental visits among occupational divers.

Study	Design / population	Main relevance for this review
Moyaux et al. (2022)	Large-scale survey	Prevalence and prevention of orofacial problems in civilian divers.
Gougeon et al. (2022)	Civilian diver study	Barodontalgia and dental barotrauma among scuba divers.
Sumen et al. (2023)	Diver study	Dental barotrauma and temporomandibular problems in divers.
García-Torres & Segura-Egea (2024)	Narrative review	Implications of barodontalgia for Navy divers and dentists.
Alolaiwi et al. (2025)	Systematic review	Prevalence and risk factors of orofacial pain among divers.
Lyons et al. (1999)	Review and experimental study	Simulated diving effects on microleakage and crown retention.
Stoetzer et al. (2012)	In vitro study	Pressure-related sealing ability of root canal obturation techniques.
Shafigh et al. (2018)	In vitro study	Pressure effects on fracture resistance and microleakage of amalgam and composite restorations.
Gulve & Gulve (2022)	In vitro study	Pressure changes and orthodontic bracket bond strength.

Source: own elaboration based on cited literature.

3.3. Clinical presentation and differential diagnosis

Clinical presentation may vary from mild transient discomfort to severe toothache that forces the diver to stop or modify diving activity. Pain may occur during descent, at depth or during ascent. The timing of pain can be diagnostically useful, but it is not sufficient alone to identify the source. A detailed history should include the phase of dive when pain occurs, depth, duration, frequency, side and tooth localization, previous dental treatment, recent restorations, endodontic treatment, sinus symptoms and temporomandibular complaints.

In odontogenic barodontalgia, the patient may identify a specific tooth or region. Pain may be sharp, throbbing, pressure-like or poorly localized. In some cases, pain disappears after return to normal pressure, which may falsely reassure the patient. A normal office examination does not exclude barodontalgia because the triggering condition may require pressure change. Therefore, dental examination should include careful inspection of restorations, percussion,

palpation, periodontal probing, vitality testing where appropriate, bite tests, radiographic evaluation and assessment for cracked tooth syndrome or recurrent caries (Robichaud & McNally, 2005; Zadik, 2009).

Differential diagnosis is essential. Maxillary sinus barotrauma or chronic sinus disease can mimic dental pain in posterior maxillary teeth. The anatomical proximity of the maxillary sinus to premolars and molars makes referred pain especially relevant in divers. Rhinologic and oral-maxillofacial diving complications have been discussed as overlapping diagnostic categories, and sinus-related causes should be considered when symptoms involve multiple posterior maxillary teeth, nasal obstruction, facial pressure or pain associated with equalization problems (Livingstone & Lange, 2018; Sionek-Wręga & Wręga, 2024).

Temporomandibular disorders represent another important diagnostic category. Scuba mouthpieces may require prolonged clenching, mandibular protrusion or unusual jaw positioning, potentially provoking masticatory muscle pain or temporomandibular joint symptoms. Studies on divers have reported temporomandibular complaints and have examined the association between diving activity and jaw pain (Koob et al., 2005; Oztürk et al., 2012; Vivacqua et al., 2024). In practice, a diver may describe orofacial pain as “tooth pain” even when the primary source is muscular or joint-related.

A practical diagnostic algorithm should therefore begin with localization and timing, then separate likely odontogenic pain from sinus-related and temporomandibular causes. If the patient reports a single tooth with recent restoration, thermal sensitivity or bite pain, dental pathology is likely. If pain is diffuse, bilateral or linked with nasal symptoms, sinus pathology should be considered. If pain is associated with jaw fatigue, preauricular pain, limitation of mouth opening or prolonged mouthpiece use, temporomandibular causes become more likely.

3.4. Dental management and treatment considerations

Management should be directed at the suspected underlying pathology. If active caries, defective restoration, pulpal inflammation, necrosis or apical periodontitis is identified, definitive treatment should be performed before further diving. Temporary measures may be appropriate for urgent symptom control, but divers should be advised that untreated pathology may recur under pressure and could interfere with safety. Because diving often takes place in remote locations, prevention and stabilization before travel are more important than emergency management after symptoms occur.

For restorative dentistry, the dentist should pay attention to marginal integrity, occlusal cracks, recurrent caries, overhangs, temporary restorations and recent cementation. Poorly retained

provisional restorations are a particular concern because pressure changes and mechanical loading from mouthpieces may contribute to dislodgement. If a diver has recently received a temporary crown, provisional bridge or temporary endodontic restoration, definitive stabilization before planned diving should be considered when possible (Gulve & Gulve, 2013a; Khanna, 2010).

Endodontic management should aim for a dense, well-sealed obturation and coronal seal. Laboratory evidence suggests that sealing ability under pressure conditions varies according to technique and material, although in vitro findings cannot be translated directly into clinical superiority in all situations (Stoetzer et al., 2012; Safai et al., 2019; von See et al., 2012). Clinically, the most important principles are elimination of infection, avoidance of residual voids, adequate temporary or permanent coronal restoration and radiographic control when indicated.

Extraction and oral surgical procedures raise separate concerns. After extraction, diving should generally be postponed until soft tissue healing is adequate and the risk of bleeding, infection, pain or sinus communication is minimized. The exact interval should be individualized according to the procedure, healing, presence of oroantral communication risk, medications and the diver's planned exposure. Although specific evidence-based waiting periods are limited, conservative counselling is justified because barometric stress, mouthpiece use and remote access to care may complicate early postoperative recovery.

Dental implants and fixed prosthodontics require evaluation of stability, occlusion and recent surgical or prosthetic stages. The evidence specific to implants and hyperbaric or diving pressure is limited, but general principles of healing and mechanical stability apply. Divers with recent implant placement, sinus lift, bone augmentation or prosthetic delivery should be advised to obtain individualized clearance from the treating dentist or surgeon before returning to diving.

Patients who develop dental pain during diving should be advised not to ignore the symptom. Repeated exposure may worsen the underlying condition or result in structural failure. If pain occurs during a dive, immediate underwater decision-making must follow diving safety training and the diver's condition; however, from a dental perspective, the patient should avoid further dives until clinical evaluation is completed. Emergency treatment should focus on identifying odontogenic infection, cracked tooth, defective restoration, sinus pathology or temporomandibular dysfunction.

3.5. Prevention and recommendations for divers

Prevention is the most practical element of dental care for divers. Routine dental check-ups should be encouraged, especially for occupational, military, technical or frequent recreational divers. Preventive visits should focus not only on caries and periodontal disease, but also on restorations, crowns, endodontically treated teeth, periapical lesions, impacted teeth, prostheses and symptoms that may suggest sinus or temporomandibular disorders (Zadik & Drucker, 2011; Onose et al., 2020; García-Torres & Segura-Egea, 2024).

Before planned intensive diving activity, dentists should ask about upcoming travel, diving depth, frequency and previous symptoms under pressure. Such questions are rarely part of routine dental history, but they can identify patients who may benefit from targeted preventive care. A diver with multiple defective restorations, untreated caries, recent endodontic treatment or recurrent sinus symptoms may require stabilization before diving. In military and occupational settings, periodic dental clearance can be integrated into fitness-to-dive assessments.

Patient education should be concise and practical. Divers should understand that tooth pain during diving may indicate dental disease even when pain disappears after surfacing. They should be advised to seek dental examination after any pressure-related dental pain, restoration loss, tooth fracture or unusual orofacial pain. They should also be cautioned against diving immediately after major dental treatment without professional advice, particularly after oral surgery, implant placement, sinus-related procedures, extensive restorative treatment or treatment involving provisional restorations.

Mouthpiece-related prevention is relevant when symptoms involve jaw fatigue or temporomandibular discomfort rather than localized dental pain. Proper fit, avoidance of excessive clenching and consideration of customized mouthpieces may help some divers, particularly those with pre-existing temporomandibular disorders or long dive durations. However, mouthpiece management does not replace dental diagnosis when symptoms are localized to a tooth (Hobson & Newton, 2001; Koob et al., 2005; Vivacqua et al., 2024).

Table 2. Practical dental recommendations for scuba divers.

Clinical situation	Dental approach	Practical recommendation
Untreated caries or defective restoration	Diagnose and restore before planned diving.	Avoid diving with active pain, open margins or suspected pulpal involvement.
Recent temporary crown or provisional restoration	Assess retention and marginal seal.	Prefer definitive stabilization before diving when feasible.
Endodontically treated tooth	Check symptoms, coronal seal and radiographic status.	Do not clear symptomatic or inadequately restored teeth for diving.
Pressure-related tooth pain with no obvious lesion	Perform targeted diagnostic tests and radiographs.	Consider cracked tooth, recurrent caries, pulp disease and referred sinus pain.
Posterior maxillary pain with nasal symptoms	Evaluate dental causes and consider ENT/sinus assessment.	Do not assume all upper molar pain is odontogenic.
Jaw fatigue or preauricular pain during diving	Screen for temporomandibular disorder and mouthpiece issues.	Consider mouthpiece modification or referral if symptoms persist.
Recent extraction or oral surgery	Assess wound healing and complications.	Postpone diving until healing is clinically satisfactory.
Professional, military or frequent diver	Include diving history in dental assessment.	Recommend periodic preventive dental examinations.

Source: own elaboration based on cited literature.

4. Discussion

This narrative review shows that barodontalgia in scuba divers is best understood as a pressure-triggered symptom with multiple possible dental and orofacial causes. For clinical dentistry, the most important message is that barodontalgia should not be dismissed when symptoms disappear after surfacing. The absence of pain at normal pressure does not exclude caries, pulpal inflammation, periapical disease, cracked tooth, defective restoration or sinus-related referred pain. This agrees with the concept that barodontalgia frequently reflects pre-existing subclinical pathology (Robichaud & McNally, 2005; Zadik, 2009; Zadik & Drucker, 2011).

The available evidence is heterogeneous. Studies differ in diver population, exposure, questionnaire design, outcome definitions and diagnostic confirmation. Military divers, occupational divers and recreational divers are not directly comparable because of differences in medical screening, diving frequency, depth, equipment, physical demands and access to preventive care. This explains why reported prevalence varies widely across studies and why meta-analysis is difficult or inappropriate in many reviews (Alwohaibi et al., 2020; Moyaux et al., 2022; Alolaiwi et al., 2025). For a clinician, however, even heterogeneous evidence is useful because it consistently identifies dental disease, restorations and diving exposure as recurring factors.

The relationship between dental restorations and pressure-related symptoms is clinically plausible but not always easy to prove in individual cases. Laboratory studies suggest that pressure cycling may influence microleakage, retention or mechanical properties of crowns, composites, amalgams, posts and orthodontic attachments (Lyons et al., 1999; Shafigh et al., 2018; Shahnavaizi et al., 2021; Ata et al., 2022; Gulve & Gulve, 2022). At the same time, laboratory pressure protocols cannot fully reproduce the complex oral environment, including occlusal loading, saliva, temperature changes, biofilm and long-term material aging. Therefore, these studies should guide clinical caution rather than serve as rigid rules.

Endodontic considerations are particularly relevant because pressure-related pain may arise from pulpal or periapical pathology and because root-filled teeth may contain microscopic spaces if treatment is inadequate. Experimental evidence suggests that obturation technique and sealing quality can influence pressure transmission and dye penetration under simulated diving conditions (Stoetzer et al., 2012; Safai et al., 2019; von See et al., 2012). The practical conclusion is not that all divers require a specific endodontic material, but that endodontic treatment in divers should aim for excellent technical quality and a durable coronal seal.

Differential diagnosis remains a key challenge. Dental pain during diving may originate from teeth, sinuses, temporomandibular joints or masticatory muscles. This is especially relevant for posterior maxillary pain, where sinus barotrauma can mimic odontogenic pain. Temporomandibular complaints may also be misinterpreted as toothache by patients, particularly when discomfort is related to prolonged mouthpiece clenching (Koob et al., 2005; Oztürk et al., 2012; Livingstone & Lange, 2018; Vivacqua et al., 2024). A structured history and examination are therefore necessary.

Compared with broad reviews of oral health complications in diving, the present review intentionally focuses on the dental approach to barodontalgia. This narrower perspective may be useful in everyday dental practice. The dentist does not need to manage all aspects of diving

medicine, but should recognize pressure-related dental symptoms, identify likely oral risk factors, treat active disease and know when referral to an ENT specialist, diving physician or temporomandibular disorder specialist is appropriate.

The review has limitations. It was narrative rather than systematic, and the selection of literature was intended to support a practical clinical synthesis. The evidence base includes older foundational papers, cross-sectional surveys, case reports and laboratory studies with limited direct comparability. Many recommendations are therefore based on expert interpretation rather than high-quality randomized or prospective data. Future research should use standardized definitions of barodontalgia and dental barotrauma, clinically verified diagnoses, consistent diving exposure variables and longitudinal designs. Studies comparing dental outcomes before and after preventive dental screening in divers would be particularly valuable.

5. Conclusions

Barodontalgia in scuba divers is a clinically relevant dental condition because it may reveal previously undetected oral pathology and may interfere with diving comfort and safety. It is most often associated with dental or orofacial risk factors such as caries, defective restorations, pulpal or periapical disease, periodontal disease, residual air spaces, sinus disorders and temporomandibular complaints.

Dentists should include diving exposure in the history of patients who report pressure-related tooth or facial pain. Diagnosis should not rely only on symptoms at normal atmospheric pressure, because pain may be provoked only during descent or ascent. A structured examination should include assessment of restorations, pulp vitality, periapical status, periodontal condition, cracked teeth, sinus-related symptoms and temporomandibular function. The most practical preventive strategy is stabilization of oral health before diving. Active dental disease, defective restorations and symptomatic endodontic or prosthetic problems should be treated before diving activity. Divers should be educated to seek dental evaluation after any pressure-related dental pain, fracture, loss of restoration or recurrent orofacial discomfort during dives.

Although current evidence remains heterogeneous, available studies support an important preventive role for dentists in recreational, occupational and military diving populations. A simple dental protocol based on risk identification, definitive treatment and patient education may reduce the likelihood of barodontalgia and dental barotrauma.

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